

Module 5

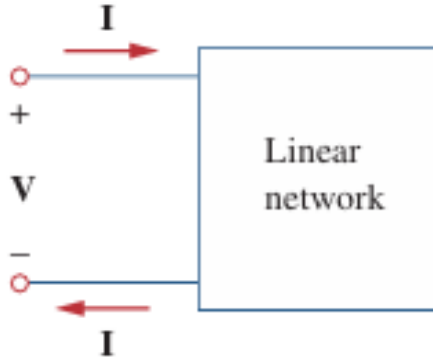
Two-Port Networks and Network Graphs

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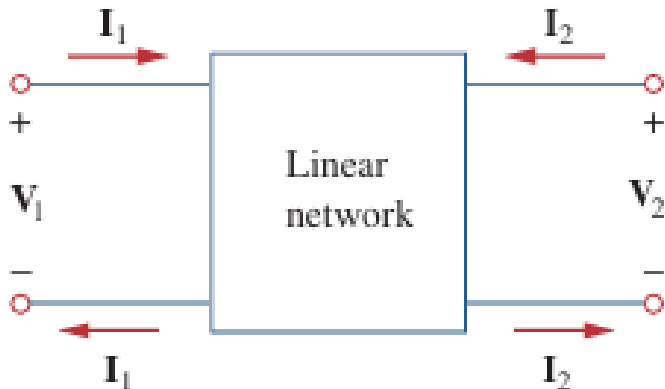
Two port networks

A pair of terminals through which a current may enter or leave a network is known as a port. Two-terminal devices or elements (such as resistors, capacitors, and inductors) result in one-port networks.



In general, a network may have n ports. A **port** is an access to the network and consists of a pair of terminals;

Thus, a two-port network has two terminal pairs acting as access points. As shown in figure, the current entering one terminal of a pair leaves the other terminal in the pair.

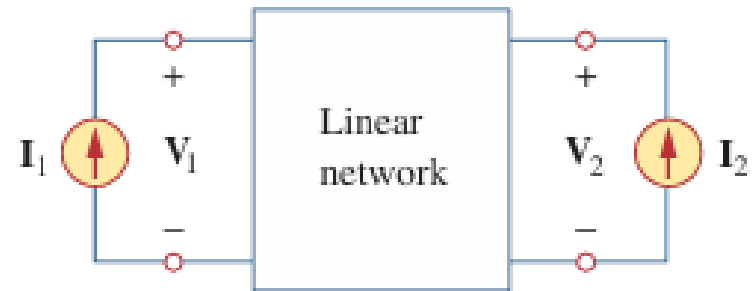
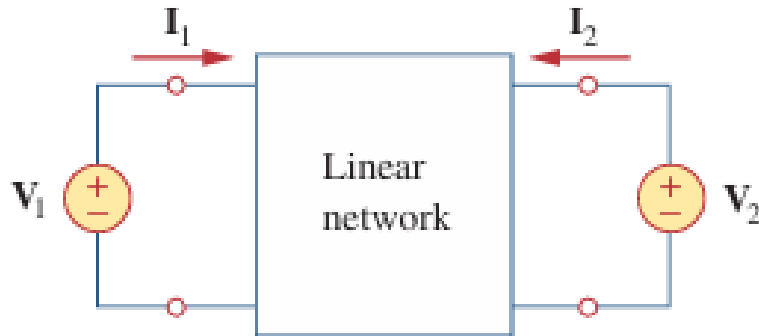


To characterize a two-port network, we require the terminal quantities V_1 , V_2 , I_1 , and I_2 , out of which two are independent.

The various terms that relate these voltages and currents are called parameters.

Impedance parameters (z-parameters)

A two-port network may be voltage-driven or current-driven.



The terminal voltages can be related to the terminal currents as:

$$\begin{aligned} V_1 &= z_{11}I_1 + z_{12}I_2 \\ V_2 &= z_{21}I_1 + z_{22}I_2 \end{aligned}$$

where the z terms are called the impedance parameters, or simply z parameters, and have units of ohms.

or in matrix form as

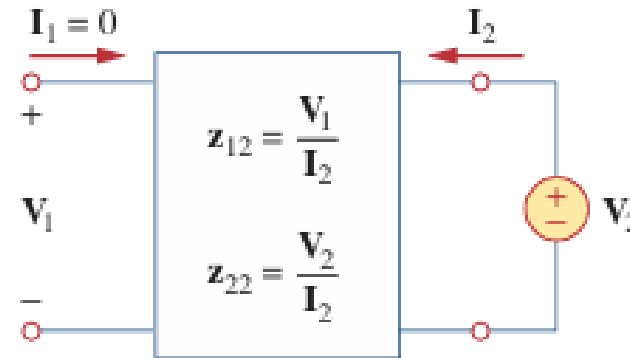
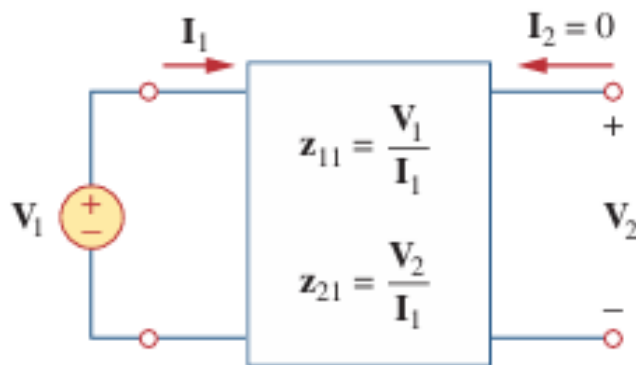
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = [z] \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

z- parameters

The values of the **z parameters** can be evaluated by setting (input port open-circuited) or (output port open-circuited). Thus,

$$\begin{aligned} z_{11} &= \left. \frac{V_1}{I_1} \right|_{I_2=0}, & z_{12} &= \left. \frac{V_1}{I_2} \right|_{I_1=0} \\ z_{21} &= \left. \frac{V_2}{I_1} \right|_{I_2=0}, & z_{22} &= \left. \frac{V_2}{I_2} \right|_{I_1=0} \end{aligned}$$

Since the **z** parameters are obtained by open-circuiting the input or output port, they are also called the **open-circuit impedance parameters**.



z_{11} = Open-circuit input impedance

z_{12} = Open-circuit transfer impedance from port 1 to port 2

z_{21} = Open-circuit transfer impedance from port 2 to port 1

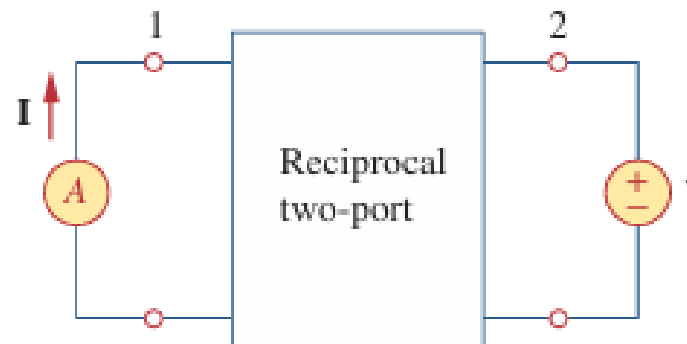
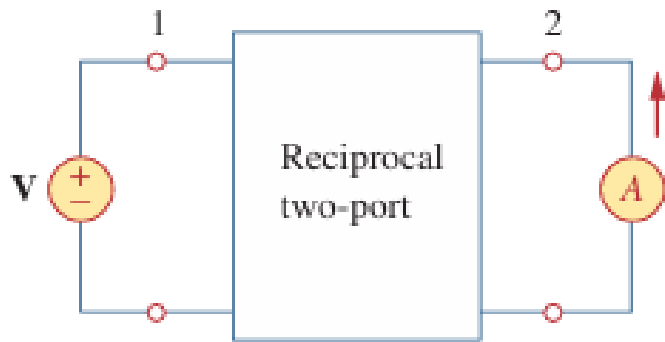
z_{22} = Open-circuit output impedance

z - parameters

z_{11} and z_{22} are called driving-point impedances, while z_{12} and z_{21} are called transfer impedances.

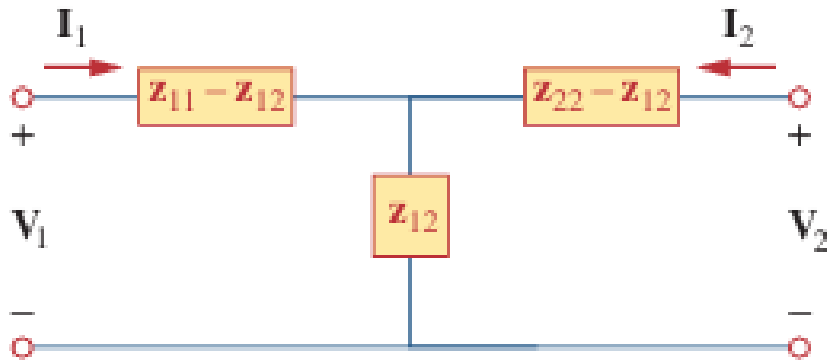
When $z_{11} = z_{22}$, the two-port network is said to be symmetrical. This implies that the **network has mirror like symmetry about some center line**; that is, a line can be found that divides the network into two similar halves.

When the two-port network is **linear and has no dependent sources**, the transfer impedances are equal ($z_{12} = z_{21}$), and the **two-port is said to be reciprocal**. This means that if the points of excitation and response are interchanged z_{12} and z_{21} remain the same.

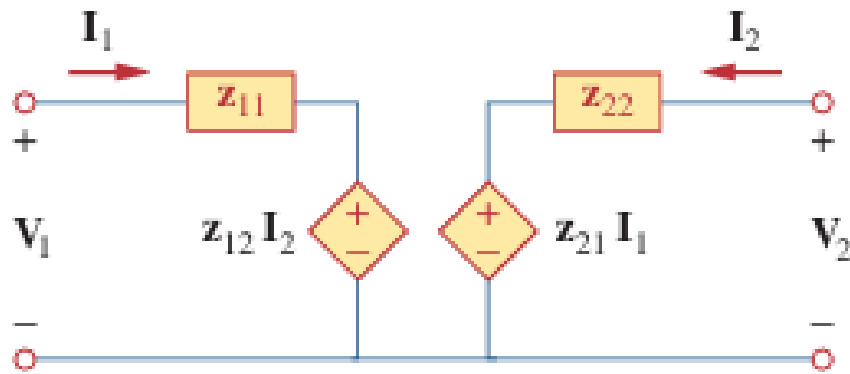


z - parameters

A reciprocal network can be replaced by the T-equivalent circuit:

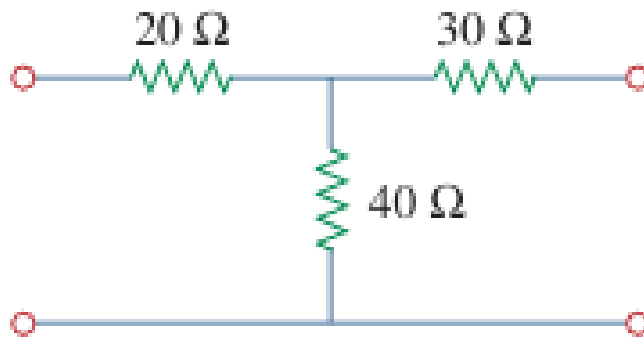


If the network is not reciprocal, a more general equivalent network as shown below:

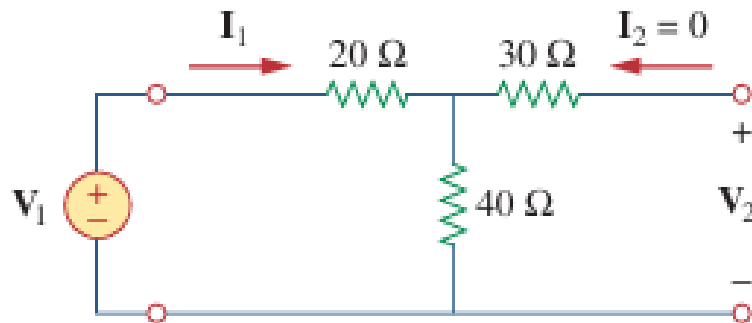


z - parameters

1. Determine the z parameters for the circuit in figure below.



To determine z_{11} and z_{21} , we apply a voltage source V_1 to the input port and leave the output port open.

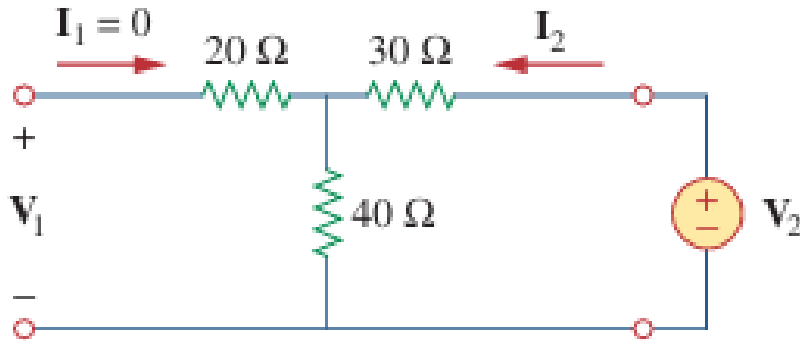


$$z_{11} = \frac{V_1}{I_1} = \frac{(20 + 40)I_1}{I_1} = 60 \, \Omega$$

$$z_{21} = \frac{V_2}{I_1} = \frac{40I_1}{I_1} = 40 \, \Omega$$

z - parameters

To find z_{12} and z_{22} , we apply a voltage source V_2 to the output port and leave the input port open.

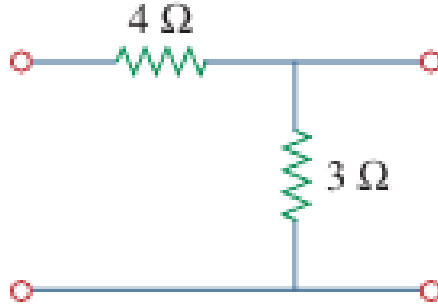


$$z_{12} = \frac{V_1}{I_2} = \frac{40I_2}{I_2} = 40\ \Omega, \quad z_{22} = \frac{V_2}{I_2} = \frac{(30 + 40)I_2}{I_2} = 70\ \Omega$$

$$[z] = \begin{bmatrix} 60\ \Omega & 40\ \Omega \\ 40\ \Omega & 70\ \Omega \end{bmatrix}$$

z - parameters

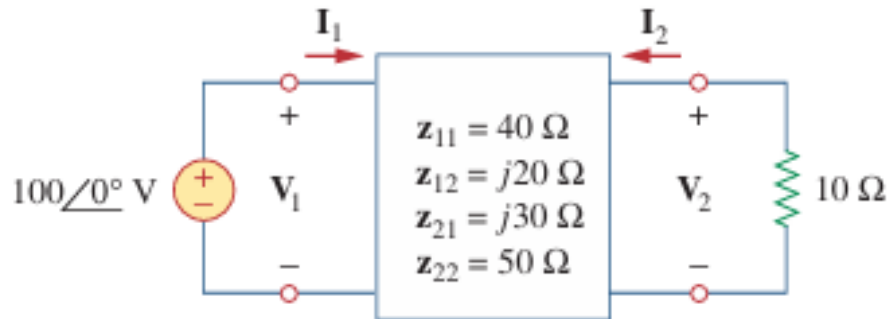
2. Find the z parameters of the two-port network in figure below.



Answer:

$$\mathbf{z}_{11} = 7 \, \Omega, \mathbf{z}_{12} = \mathbf{z}_{21} = \mathbf{z}_{22} = 3 \, \Omega.$$

3. Find I_1 and I_2 in the circuit.



This is not a reciprocal network.

$$\mathbf{V}_1 = 40\mathbf{I}_1 + j20\mathbf{I}_2$$

$$\mathbf{V}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2$$

z - parameters

$$\mathbf{V}_1 = 40\mathbf{I}_1 + j20\mathbf{I}_2$$

$$\mathbf{V}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2$$

$$\mathbf{V}_1 = 100\angle 0^\circ, \quad \mathbf{V}_2 = -10\mathbf{I}_2$$

$$100 = 40\mathbf{I}_1 + j20\mathbf{I}_2$$

$$-10\mathbf{I}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_1 = j2\mathbf{I}_2$$

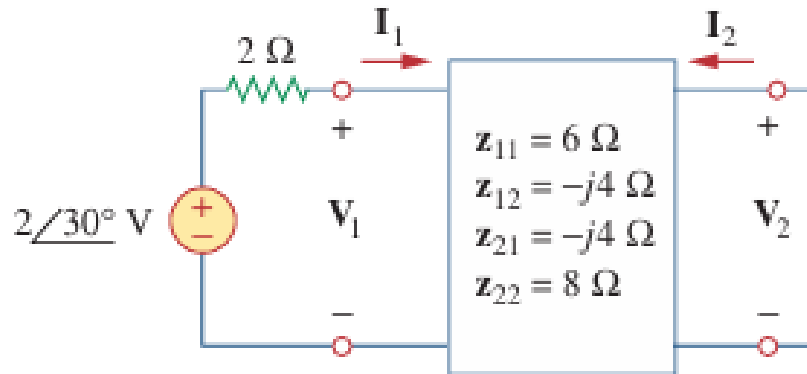
$$100 = j80\mathbf{I}_2 + j20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = \frac{100}{j100} = -j$$

$$\mathbf{I}_1 = j2(-j) = 2$$

$$\mathbf{I}_1 = 2\angle 0^\circ \text{ A}, \quad \mathbf{I}_2 = 1\angle -90^\circ \text{ A}$$

z - parameters

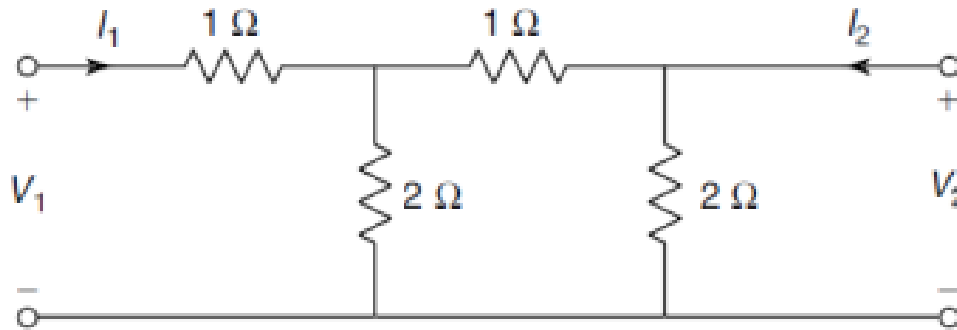
4. Calculate and in the two-port.



Answer:

$$200 \angle 30^\circ \text{ mA}, 100 \angle 120^\circ \text{ mA}.$$

5. Find z-parameters for the network shown.

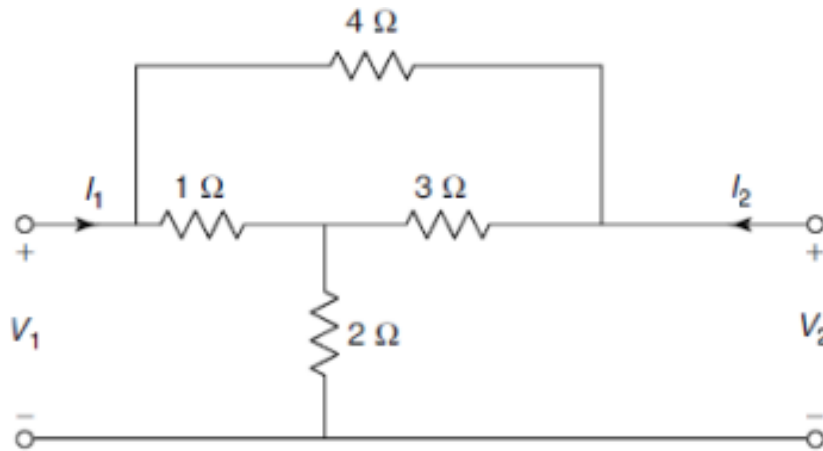


Answer:

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{11}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{6}{5} \end{bmatrix}$$

z - parameters

6. Find the open-circuit impedance parameters for the network shown in Fig. Determine whether the network is symmetrical and reciprocal.



Answer:

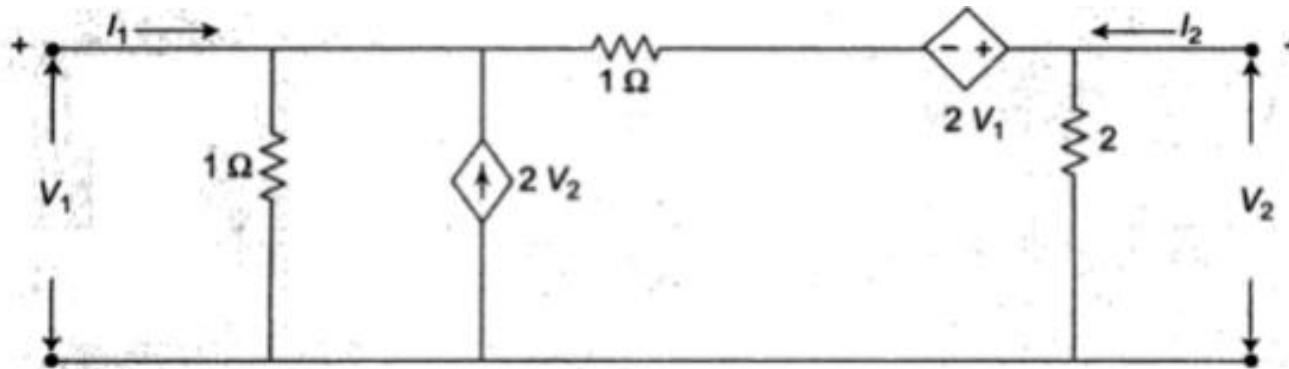
$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{23}{8} & \frac{19}{8} \\ \frac{19}{8} & \frac{31}{8} \end{bmatrix}$$

$Z_{11} \neq Z_{22}$, the network is not symmetrical.

$Z_{12} = Z_{21}$, the network is reciprocal.

z - parameters

7. Find Z_{21} and Z_{22} for the network shown BELOW:



Answer:

$$Z_{21} = \frac{-1}{2}; Z_{22} = \frac{-1}{3}$$

Next class: Admittance parameters